Tempered Langevin diffusions with heavy-tailed targets: explosions, moment thresholds and ergodicity

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Part 2: Proofs

TLDs with polynomially decaying targets

Let Π be a probability measure on $\mathbb{R}^d$ and $\Pi(\mathrm{d}x)/\mathrm{d}x \propto \pi(x)$ where $\pi \in C^2(\mathbb{R}^d)$ satisfies

$$|x|^{d+r}\pi(x) \to c_\pi > 0 \quad \text{and} \quad \langle x, \nabla \log \pi(x)\rangle \to -(d+r) \quad \text{as} \quad |x| \to \infty^1$$

for some $r > 0$. The *tempered Langevin diffusion* X with temperature $\ell \geq 0$ solves the SDE

$$\mathrm{d}X_t = (1-2\ell)\pi(X_t)^{-2\ell}\nabla \log \pi(X_t)\mathrm{d}t + \sqrt{2}\pi(X_t)^{-\ell}\,\mathrm{I}_d\,\mathrm{d}B_t$$

with stationary measure Π. As shown in [FR05; BM26], we have



Question: which property of X_t drives this transition in convergence rates?

[1] For $f : \mathbb{R}^d \to \mathbb{R}$, we write $f(x) \to c \in \mathbb{R}$ as $|x| \to \infty$ if $\lim_{\rho\to\infty} \sup_{|x|=\rho} |f(x) - c| = 0$.

Main results: explosions, moment thresholds and ergodicity

Assumption: $|x|^{d+r}\pi(x) \to c_\pi > 0$ and $\langle x, \nabla \log \pi(x)\rangle \to -(d+r)$ as $|x| \to \infty$.

	Explosion	**Moment thresholds of $\lvert X_t\rvert^p$ $(p>0)$**	**Ergodicity in f_p-variation & L^p-Wasserstein**
$0 \le \ell < \frac{1}{d+r}$	$\zeta = \infty$ a.s.	$\forall t>0 : \mathbb{E}_x[e^{\lvert X_t\rvert^p}] < \infty \iff p \in (0, 2(1-\ell(d+r)))$	$\approx t^{-\frac{r-p}{2(1-\ell(d+r))}}\ \forall\, p \in (0,r)$
$\ell = \frac{1}{d+r}$	$\zeta = \infty$ a.s.	$\forall t>0 : \mathbb{E}_x[\lvert X_t\rvert^p] < \infty$, but $\mathbb{E}_x[e^{\lvert X_t\rvert^p}] = \infty$	Exponential $\forall\, p \in (0,r)$, but not uniform
$\frac{1}{d+r} < \ell < \frac{1+r/2}{d+r}$	$\zeta = \infty$ a.s.	$\mathbb{E}_x[\lvert X_t\rvert^p] < \infty \iff p \in (0,r)$	Uniform $\forall\, p \in (0,r)$
$\ell = \frac{1+r/2}{d+r}$	$\zeta = \infty$ a.s.	$\mathbb{E}_x[\lvert X_t\rvert^p] < \infty \iff p \in (0,r)$	Uniform $\forall\, p \in (0,r)$
$\frac{1+r/2}{d+r} < \ell < \infty$	$\zeta < \infty$ a.s.	\	\

Ergodicity decay rates: $\|\mathbb{P}_x(X_t \in \cdot) - \Pi(\cdot)\|_{f_p}$, for $f_p(x) = 1 + |x|^p$, and $\mathcal{W}_p(\mathbb{P}_x(X_t \in \cdot), \Pi)^{p \vee 1}$, for $\boldsymbol{p > 0}$ (since $r > 0$), where $\mathcal{W}_p(\mu,\nu) = \inf_{\eta \in C(\mu,\nu)} (\int_{\mathbb{R}^d \times \mathbb{R}^d} |x-y|^p \, \eta(\mathrm{d}x, \mathrm{d}y))^{1/(p\vee 1)}$.

Explosions

Explosion time: $\ell \neq \frac{1+r/2}{d+r}$

We apply **Khasminskii's explosion test** [Kha12, Thm 3.5] to prove that

$$X \text{ is non-explosive if } 0 \leq \ell < \frac{1+r/2}{d+r} \quad \text{and} \quad \zeta < \infty \ \mathbb{P}_x\text{-a.s. for } x \in \mathbb{R}^d \text{ if } \ell > \frac{1+r/2}{d+r}.$$

- $\mathbf{0 \leq \ell < \frac{1+r/2}{d+r}}$: we construct $\Psi_+(t) := \int_1^t \Big(\frac{d-1}{s} + \sup_{|x|=s} \frac{(1-2\ell)\langle \nabla \log \pi(x), x\rangle}{s} \Big) \mathrm{d}s$ and

  $$U_+(x) := 1 + \int_1^{|x|} e^{-\Psi_+(s)} \mathrm{d}s, \quad \text{so that } \lim_{\rho \to \infty} \inf_{|x|=\rho} U_+(x) = \infty \text{ and } \mathcal{A}U_+ \leq KU_+, \ K \geq 0.$$

- $\boldsymbol{\ell > \frac{1+r/2}{d+r}}$: we define Ψ_- in the same way as Ψ_+ with sup replaced by inf and

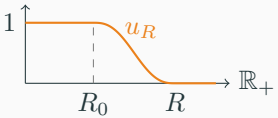
  $$U_-(x) := \int_0^{|x|} e^{-\Psi_-(\rho)} \left(\int_0^\rho e^{\Psi_-(t)} \sup_{|y|=t} \pi(y)^{2\ell} \, \mathrm{d}t \right) \mathrm{d}\rho \quad \text{for} \quad x \neq 0,$$

  so that U_- is bounded and $\mathcal{A}U_- \geq c > 0$.

Non-explosion at $\ell = \frac{1+r/2}{d+r}$

- **Cutoff function**: $\zeta_R := \inf\{t > 0 : |X_t| > R\}$. For $T > 0$ and every $x \in \mathbb{B}_{R_0}$,

$$\mathbf{1}_{\{\zeta_R \le T\}} \le \sup_{t\in[0,T]} |u_R(|X_t|) - u_R(|X_0|)|^2 \quad \mathbb{P}_x\text{-a.s. where}$$



- Y **diffuses like** X **in** $\mathbb{B}_{2R}$ **with co-normal reflection at** $\partial\mathbb{B}_{2R}$. Hence Y is Π_R-reversible where

$$\Pi_R(\mathrm{d}x)/\mathrm{d}x \propto \pi(x)\mathbf{1}_{\{x\in\mathbb{B}_{2R}\}}/Z_R \quad \text{and} \quad Z_R = \int_{\mathbb{B}_{2R}} \pi(x)\mathrm{d}x.$$

- **Itô trick** [LZ94]: $(\widehat{Y}_t)_{t\in[0,T]} := (Y_{T-t})_{t\in[0,T]}$ has the same law as $(Y_t)_{t\in[0,T]}$ under $\mathbb{P}_{\Pi_R}$ and

$$u_R(|Y_t|) - u_R(|Y_0|) = \frac{1}{2}M_t - \frac{1}{2}\Big(\widehat{M}_T - \widehat{M}_{T-t}\Big)$$

where $(M_t)_{t\in[0,T]}$ and $(\widehat{M}_t)_{t\in[0,T]}$ are the martingale parts of $u_R(|Y_t|)$ and $u_R(|\widehat{Y}_t|)$.

Moment thresholds

$\ell \in \left[0, \frac{1}{d+r}\right)$: $\mathbb{E}_x[e^{|X_t|^p}] < \infty$ **for all** $t > 0 \iff p \in (0, 2(1-\ell(d+r)))$

Consider a C^2-function $\Phi : \mathbb{R}^d \to (0,\infty)$ s.t. $\Phi(x) = |x|^{-\alpha}/\left(\sqrt{2}|\alpha|\right)$ for sufficiently large $|x|$ where $\alpha = \ell(d+r) - 1 \in [-1, 0)$. By Itô's formula,

$$\Phi(X_t) = \Phi(x) + \int_0^t \mathcal{A}\Phi(X_s)\mathrm{d}s + M_t \quad \text{with} \quad \langle M\rangle_t = \int_0^t 2\pi(X_s)^{-2\ell}|\nabla\Phi(X_s)|^2\mathrm{d}s.$$

Since $\mathcal{A}\Phi(x) \sim -\frac{r-\alpha}{\sqrt{2}}|x|^\alpha$ and $2\pi(x)^{-2\ell}|\nabla\Phi(x)|^2 \to 1$ as $|x| \to \infty$, we thus have

$$K = \sup_{\mathbb{R}^d}|\mathcal{A}\Phi| < \infty \quad \text{and} \quad K' = \sup_{\mathbb{R}^d} 2\pi^{-2\ell}|\nabla\Phi|^2 \in (0,\infty).$$

DDS Brownian motion W on $\mathbb{R}$ [RY99, Chap. V, Thm 1.7] gives $M_t = W_{\langle M\rangle_t}$, and thus

$$\Phi(x) - Kt + W_{\langle M\rangle_t} \le \Phi(X_t) \le \Phi(x) + Kt + W_{\langle M\rangle_t} \quad \text{and} \quad \langle M\rangle_t \le K't \quad \mathbb{P}_x\text{-a.s.}$$

$\ell = \frac{1}{d+r}$: analogous bounds with $\Phi(x) = \log|x|/\sqrt{2}$ imply loss of all exponential moments.

$\ell \in \left(\frac{1}{d+r}, \frac{1+r/2}{d+r}\right]$: finite p-th moments for $0 \le p < r$ (Direct methods fail!)

Consider a C^2-function $\Phi : \mathbb{R}^d \to (0,\infty)$ s.t. $\Phi(x) = |x|^{-\alpha}/\big(\sqrt{2}|\alpha|\big)$ for large $|x|$, then

$\Phi(X) \approx$ a Bessel process of dimension $\delta_\star := r/\alpha \ge 2$ and $\mathbb{E}_x[|X_t|^p] \approx \mathbb{E}_{\Phi(x)}\big[\Phi(X_t)^{-\nu}\big]$,

where $\alpha = \ell(d+r) - 1 \ge r/2$, $p \in (0,r)$ and $\nu = p/\alpha \in (0,\delta_\star)$. However:

- **Path comparison with Bessel fails**. For a Bessel Z of dimension $\delta > 2$, $Z_0 > 0$:

$$\mathbb{E}[Z_t^{-\eta}] < \infty \iff \eta \in [0,\delta), \quad \text{but} \quad \mathbb{E}[(\inf_{s\in[0,t]} Z_s)^{-\eta}] < \infty \iff \eta \in [0, \delta-2).$$

- **Grönwall-type inequality**: it suffices to find $\mathcal{U} \approx |x|^p$ s.t. $\mathcal{A}\mathcal{U} \le c\mathcal{U}$ for $c \ge 0$. But,

$$\mathcal{A}|x|^p \ge c'|x|^{p+2\alpha} \quad \text{if} \quad p \in (r-2\alpha, r),$$

for some constant $c' > 0$ and $|x|$ large enough.

$\ell \in \left(\frac{1}{d+r}, \frac{1+r/2}{d+r}\right]$: time-dependent Lyapunov function $\mathcal{U}$ works!

Let Z be a Bessel process of dimension $\delta > 0$ and define

$$U_{\delta,\eta}(s,z) := \mathbb{E}\big[Z_s^{-\eta}\big] \quad \text{for} \quad Z_0 = z \quad \text{and} \quad \eta \in (0,\delta).$$

Recall $\nu := \frac{p}{\alpha} \in (0, \delta_\star)$. We expect $U_{\delta,\nu}(t, \Phi(x))$ solves $\partial_t U = \mathcal{A}U$, because for any $\varphi \in C^2$,

$$\mathcal{A}(\varphi \circ \Phi)(x) = \theta_\ell(x)^2 \left[\frac{\delta_\ell(x) - 1}{2z}\varphi'(z) + \frac{1}{2}\varphi''(z)\right] \quad \text{with} \quad z = \Phi(x),$$

where $\theta_\ell(x) \to 1$ and $\delta_\ell(x) \to \delta_\star$ as $|x| \to \infty$. Let $T, \varepsilon > 0$ and $\delta' \in (\nu, \delta_\star)$. We define

$$\mathcal{U}(t,x) := U_{\delta',\nu}(T - t + \varepsilon, \Phi(x)) + K(T-t), \quad K > 0.$$

There exists $K > 0$ independent of T and ε s.t.

$$(\partial_t + \mathcal{A})\,\mathcal{U} \le 0 \quad \text{on} \quad \mathbb{R}^d \quad \text{and} \quad U_{\delta',\nu}(t, \Phi(x)) \asymp (t + |x|^{-2\alpha})^{-p/(2\alpha)} \quad \text{as} \quad |x| \to \infty.$$

Ergodicity

Non-uniform ergodicity at $\ell = \frac{1}{d+r}$

- **Supermartingale**: taking $V \in C^2$ with $V(x) = \log|x|$ for large $|x|$, we have $\mathcal{A}(1/V) \le K(1/V)^2$ on $\mathbb{R}^d$ for some $K > 0$, so

$$\left(1/V(X_t) - K\int_0^t V(X_s)^{-2}\mathrm{d}s\right)_{t\ge 0} \text{ is a supermartingale.}$$

  Moreover, $\limsup_{t\to\infty} V(X_t) = \infty$ $\mathbb{P}_x$-almost surely for every $x \in \mathbb{R}^d$.

- **Lower bound on escape probability**: define $\tau_R = \inf\{s \ge 0 : V(X_s) < R\}$ for large $R > 0$. By [BM26, Lemma 4.2], for any $t > 0$ and $q \in (0,1)$,

$$q \le \mathbb{P}_x(\tau_R \ge t) \le \mathbb{P}_x\big(|X_t| \ge e^R\big) \quad \text{for} \quad \log|x| = \frac{2(2Kt+R)}{(1-q)^2}.$$

- **Lower bound on TV**: choose $R = R_q$ with $\Pi(\mathbb{B}_{e^{R_q}}) \ge q$. Then

$$\|\mathbb{P}_x(X_t \in \cdot) - \Pi\|_{\mathrm{TV}} \ge 2\big(\Pi(\mathbb{B}_{e^{R_q}}) - \mathbb{P}_x\big(|X_t| < e^{R_q}\big)\big) \ge 2(2q-1) \to 2 \quad \text{as} \quad q \uparrow 1.$$

Uniform ergodicity for $\ell \in \left(\frac{1}{d+r}, \frac{1+r/2}{d+r}\right]$ in f_p-variation and L^p-Wasserstein

To obtain uniform convergence in f_p-variation, $f_p(x) = 1 + |x|^p$ and $p \in (0, r)$, we combine

$$\int_{\mathbb{R}^d} |x|^p \Pi(\mathrm{d}x) < \infty \quad \text{and} \quad \sup_{t \geq 1} \sup_{x \in \mathbb{R}^d} \mathbb{E}_x[|X_t|^p] < \infty \quad \text{for every } p \in (0, r),$$

together with X is uniformly ergodic in total variation [FR05, Thm 16].

Fix $p \in (0, r)$ and $\tilde{p} \in (p, r)$. Take $|g| \leq f_p$ and split at radius $R \geq 1$. Inside $\mathbb{B}_R$,

$$\left| \int_{\mathbb{R}^d} g \mathbf{1}_{\mathbb{B}_R} \left(\mathbb{P}_x(X_t \in \cdot) - \Pi \right) \right| \leq 2R^p \left\| \mathbb{P}_x(X_t \in \cdot) - \Pi(\cdot) \right\|_{\mathrm{TV}} \leq 2CR^p e^{-ct}$$

for some $C, c > 0$ and all $t \geq 1$. Outside $\mathbb{B}_R$, we have $|g| \leq f_p \leq 2R^{p-\tilde{p}} f_{\tilde{p}}$, thus, for $C' > 0$,

$$\left| \int_{\mathbb{R}^d} g \mathbf{1}_{\mathbb{B}_R^c} \left(\mathbb{P}_x(X_t \in \cdot) - \Pi \right) \right| \leq 2R^{p-\tilde{p}} \left(\mathbb{E}_x[f_{\tilde{p}}(X_t)] + \int f_{\tilde{p}} \, \mathrm{d}\Pi \right) \leq 2C' R^{p-\tilde{p}}.$$

Choosing $R = e^{ct/\tilde{p}}$ yields $\left\| \mathbb{P}_x(X_t \in \cdot) - \Pi \right\|_{f_p} \leq \tilde{C} e^{-c(1-p/\tilde{p})t}$ for some $\tilde{C} > 0$ and $t \geq 1$.

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